Sequential Counterfactual Risk Minimization

Causal Inference and Missing Data Group Presentation

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Personal Presentation

- Status: last year PhD student at Criteo (with Eustache Diemert) and INRIA Thoth (with Julien Mairal, Pierre Gaillard), finishing this summer.
- PhD research: sequential and statistical learning in policy learning problems.
- Long term research ambition: effective and efficient decision making in causal inference problems with applications in healthcare.

Outline

1. Motivations
2. Counterfactual Risk Minimization
3. Sequential Designs
4. Sequential Counterfactual Risk Minimization Guarantees
5. Empirical Evaluation

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Context

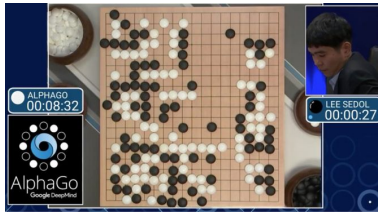
Problem: In interactive systems, full information is not always available.





We wish to learn to take "better" actions so as to optimize a feedback signal...





... while ensuring safe/affordable learning for real-life applications.

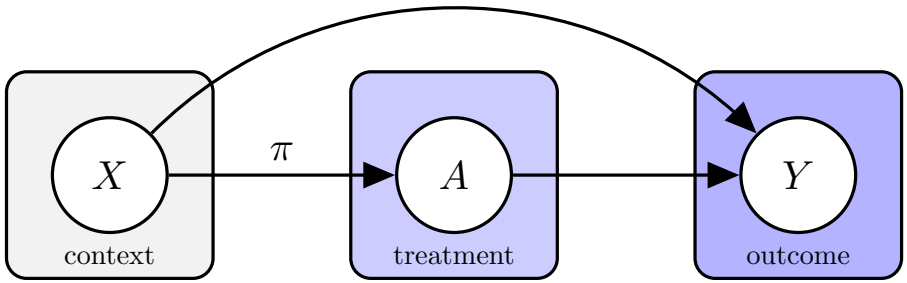




Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$ obtained from interactions of a decision-making system. In each interaction, the system:

- observes a context $x_i \in \mathcal{X}$, and $x_i \sim \mathcal{P}_{\mathcal{X}}$,
- samples an action (treatment) $a_i \in \mathcal{A}$ from a logging policy π_{θ_0} with a probability $\pi_{i,0} = \pi_{\theta_0}(a_i|x_i)$ defined by a model $\theta_0 \in \Theta$,
- observes a loss (outcome) $y_i \sim \mathcal{P}_{\mathcal{Y}}(\cdot|x_i, a_i)$



The risk L of a model $\theta \in \Theta$ (that induces a policy π_θ) is given by:

$$L(\theta) = \mathbb{E}_{x\sim\mathcal{P}_{\mathcal{X}}}\, \mathbb{E}_{a\sim\pi_\theta(\cdot|x)}\, \mathbb{E}_{y\sim\mathcal{P}_{\mathcal{Y}}(\cdot|x,a)}\, [y]$$

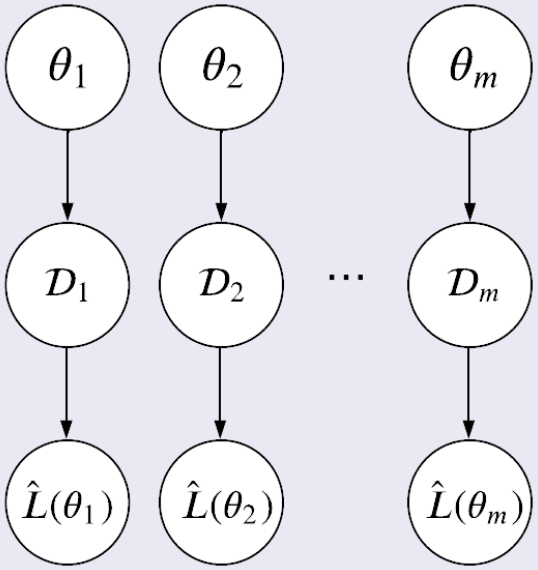
Counterfactual question: *"What would have been the risk of the system, should the model θ_0 had been θ?"*

Online versus Offline Policy Estimation

To answer the last question, the goal is to estimate $L(\theta)$ with $\hat{L}(\theta)$.

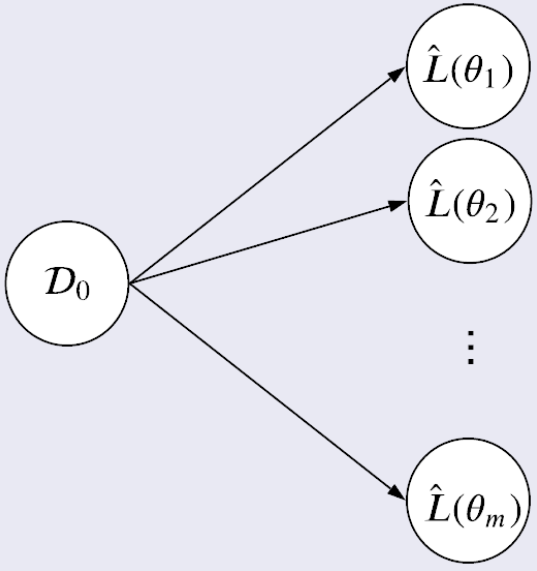
Online

A/B Tests: easy evaluation $\hat{L}$ for any θ_i but requires costly generation of new $\mathcal{D}_i$



Offline

Safe evaluation of any θ_i using only $\mathcal{D}_0$ but requires difficult estimator $\hat{L}$



In real-life applications as advertising, personalized medicine, clinical trials, offline learning may be preferred.

Counterfactual Policy Evaluation

Fortunately, it is possible to provide an answer to the counterfactual question, under further assumptions on the data generating process.

Assumption (Unconfoundedness)

$y \perp\!\!\!\perp a|x$, the treatment is exogenous and its assignment depends only on x

This assumption needs to be taken into account in the original logging system, otherwise the inference is biased (Simpson paradox).

Assumption (Common support)

The policies π_θ for $\theta \in \Theta$ have all the same support.

Under these two assumptions, it is possible to use the IPS method (Horvitz et al., 1952)

$$L(\theta) = \mathbb{E}_{x\sim\mathcal{P}_\mathcal{X}}\, \mathbb{E}_{a\sim\pi_\theta(\cdot|x)}\, \mathbb{E}_{y\sim\mathcal{P}_\mathcal{Y}(\cdot|x,a)} \left[y \frac{\pi_\theta(a|x)}{\pi_{\theta_0}(a|x)} \right]$$

Then, a natural estimator is

$$\hat{L}^{\text{IPS}}(\theta) = \frac{1}{n} \sum_{i=1}^{n} y_i \frac{\pi_\theta(a_i|x_i)}{\pi_{0,i}} \text{ where } \pi_{i,0} = \pi_{\theta_0}(a_i|x_i). \qquad \text{(IPS)}$$

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Counterfactual Risk Minimization (CRM) (Swaminathan et al., 2015)

Importance weighting has large variance and is subject to various overfitting phenomena, to control the variance a clipped estimator has been proposed:

$$\hat{L}^{\text{cIPS}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{y_i \min\left\{\frac{\pi_\theta(a_i|x_i)}{\pi_{0,i}}, \alpha\right\}}_{\nu_i(\theta)}, \qquad \text{(cIPS)}$$

where α is a clipping parameter. We can then define the sample variance penalized (Maurer et al., 2009) objective:

$$\mathcal{L}(\theta) = \hat{L}^{\text{cIPS}}(\theta) + \lambda\sqrt{\frac{\widehat{\text{Var}}(\nu_i(\theta))}{n}} \quad \text{with} \quad \lambda > 0.$$

CRM Goal: find a model θ that minimizes a regularized estimate of $L(\theta)$.

$$\hat{\theta}^{\text{CRM}} \in \arg\min_{\theta\in\Theta} \mathcal{L}(\theta) \qquad \text{(CRM)}$$

The underexploration problem

When the logging policy underexplored the action space, it is hard to perform offline evaluation due to the IPS variance issues. Assessing with high statistical significance the quality of policies that sample actions in underexplored areas is difficult.

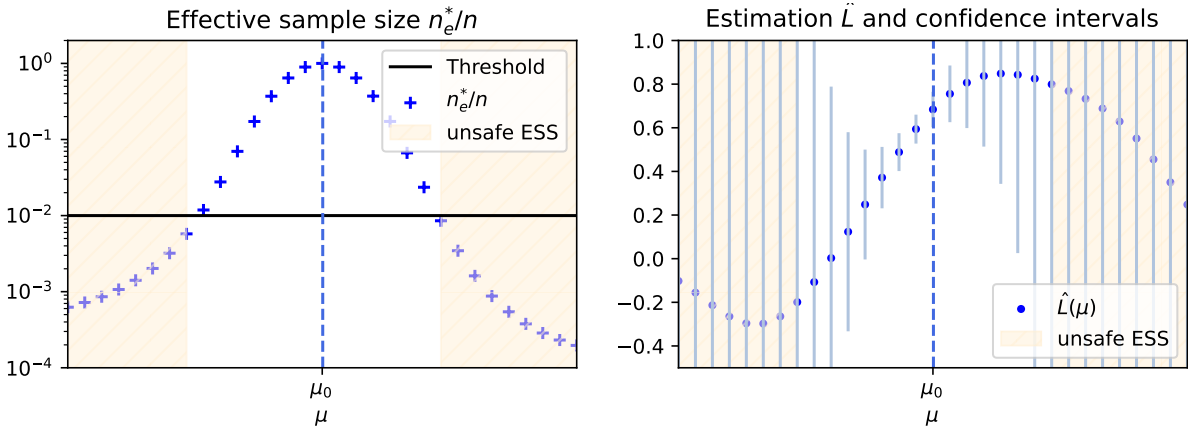


Figure: Example of importance sampling diagnostics and confidence interval for counterfactual evaluation of Gaussian distributions w.r.t a logging parameter μ_0.

Therefore it is sometimes necessary to increase the sample size by collecting additional samples with *sequential* deployments.

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Sequential Designs

The idea is to *adaptively* collect data instead of redeploying the same logging model θ_0

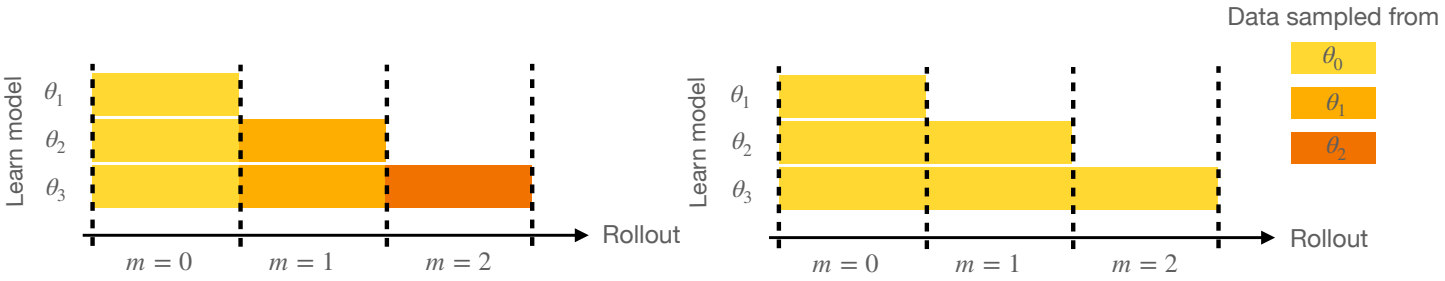


Figure: Sequential designs (left) and non-adaptive redeployment (right).

At a round m, a model $\theta_m \in \Theta$ is deployed and a new set s_m of n_m observations $s_m = (x_{m,i}, a_{m,i}, y_{m,i})_{i=1,\dots,n_m}$ is collected to define an objective function $\mathcal{L}_m$ and learn a new model θ_{m+1} and reiterate.

Algorithm Sequential Counterfactual Risk Minimization

Input: Logged observations $(x_{0,i}, a_{0,i}, y_{0,i}, \pi_{0,i})_{i=1,\dots,n_0}$, parameter $\lambda > 0$
for $m = 0$ *to* $M - 1$ **do**
  Build $\mathcal{L}_m$ from observations s_m
  Learn θ_{m+1}
  Deploy the model θ_{m+1} and collect observations s_{m+1}
end

Sequential Counterfactual Risk Minimization

We extend the (CRM) learning principle to *sequential* learning (SCRM) principle.

- We introduce a **novel IPS-IX** counterfactual estimator with **improved variance-dependent** convergence guarantees.
- This allows us to **improve the performance of CRM** in terms of excess risk and regret rates, by using an analysis similar to **restart strategies** in accelerated optimization methods.
- We also provide an **empirical evaluation** of our method in both discrete and continuous action settings that demonstrates the **practical benefits of SCRM**.

Houssam Zenati, Eustache Diemert, Matthieu Martin, Julien Mairal, and Pierre Gaillard (2023). "Sequential Counterfactual Risk Minimization". In: URL: https://arxiv.org/abs/2302.12120

Illustrating the benefits of sequential designs

Let us consider Gaussian parametrized policies $\pi_\theta = \mathcal{N}(\theta, \sigma^2)$ and a loss $l_t(a) = (a - y_t)^2 - 1$ where $y_t \sim \mathcal{N}(\theta^*, \sigma^2)$. We illustrate in Figure 3 the evolution of the losses of learned models θ_m with either:

- Batch CRM learning with redeployment of θ_0
- Sequential CRM learning with models $\theta_0, \ldots, \theta_{m-1}$ deployed adaptively

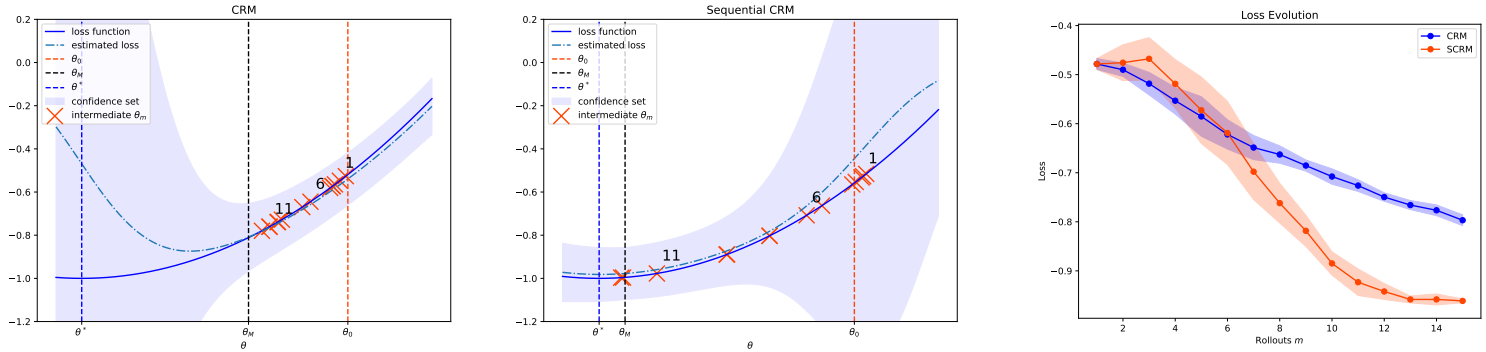


Figure: Comparison of CRM (left) and SCRM (middle) on a simple setting. Models learned $\theta_1, \ldots, \theta_M$ indicated with 'x' markers and rollout numbers. Evolution of averaged losses (right)

We see that the models learned with SCRM take larger optimization steps than the ones with CRM.

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IPS-IX: Implicit exploration and controlled variance

We propose a new counterfactual estimator. Denoting $\pi_{\theta,i} = \pi_\theta(a_{m,i}|x_{m,i})$ and introducing with $\alpha > 0$, we change the importance weighted loss:

$$\begin{cases} \nu_i(\theta) & = y_{m,i} \min\left\{\dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i}}, \alpha\right\} & \text{(cIPS)} \\ \zeta_{i,m}(\theta) & = y_{m,i}\left(\dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i} + \alpha\pi_{\theta,i}} - 1\right) & \text{(IPS-IX)} \end{cases}$$

Proposition (Generalization Error Bound, informal)

Let $\theta \in \Theta$, then with high propability:

$$L(\theta) \lesssim \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{Var}(\zeta_{i,m}(\theta))}{n_m}} + \frac{\lambda_m^2}{n_m} + \delta_m,$$

for some δ_m and a λ_m depending on a complexity measure of Θ.

IPS-IX resembles the implicit exploration estimator in the EXP3-IX algorithm (Lattimore et al., 2020).

Variance-dependent convergence guarantees

The latter motivates to solve for $m \in \{1, \dots M\}$:

$$\hat{\theta}_{m+1} \in \arg\min_{\theta \in \Theta} \mathcal{L}_m(\theta), \tag{SCRM}$$

where $\mathcal{L}_m(\theta) = \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{\text{Var}}(\zeta_{i,m}(\theta))}{n_m}}$. Let $\theta^* \in \arg\min_\theta L(\theta)$.

Proposition (Excess Risk Bound, informal)

With high probability:

$$L(\theta_{m+1}) - L(\theta^*) \lesssim \sqrt{\nu_m^2 \frac{\log n_m}{n_m}},$$

where $\nu_m^2 = \text{Var}_{x,\theta_m}\left(\frac{\pi_{\theta^*}(a|x)}{\pi_{\theta_m}(a|x)}\right)$.

IPS-IX improves the dependency in ν_m^2, compared to $\nu_m^2 + 1$ in previous work (Zenati, Bietti, et al., 2020).

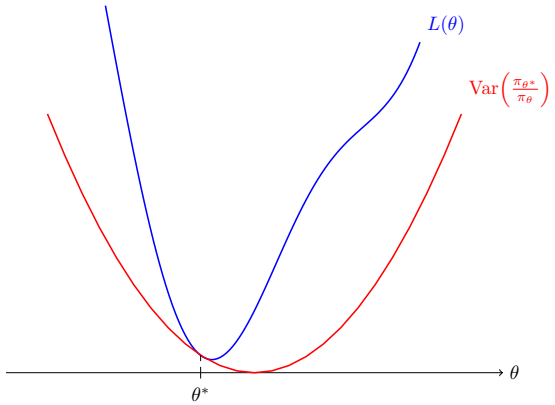
This turns out to be a crucial point to use these error bounds sequentially as in acceleration methods since $\nu_m \to 0$ if $\theta_m \to \theta^*$.

SCRM analysis

Assumption (Hölderian Error Bound)

There exist $\gamma > 0$ *and* $\beta > 0$ *such that for any* $\theta \in \Theta$,

$$\gamma \mathrm{Var}_{x,\theta}\left(\frac{\pi_{\theta^*}(x|a)}{\pi_\theta(x|a)}\right) \leq (L(\theta) - L(\theta^*))^\beta \ .$$



In acceleration methods, Hölderian error bounds are of the form $\gamma d(\theta, \theta^*) \leq (L(\theta) - L(\theta^*))^\beta$ with d a distance.

Proposition (SCRM Excess Risk and Regret, informal)

Let $n_0, n \geq 2$, $n_m = n_0 2^m$ *for* $m = 0, \ldots, M \approx \left\lfloor \log \frac{n}{n_0} \right\rfloor$.

$$L(\theta_M) - L(\theta^*) \leq O\left(n^{-\frac{1}{2-\beta}} \log n\right).$$

We obtain fast rates on par with (Bartlett et al., 2006; Van Erven et al., 2015) and when $\beta = 0$, we recover the CRM rates (Zenati, Bietti, et al., 2020).

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IPS-IX in practice

We simulate the setting where the logging policy underexplores the action space and consider sinusoidal losses $y(a) = \cos(a)$ and evaluated policies are $\{\pi_i = \mathcal{N}(i * \pi/4, 1)\}_{i=0,4}$.

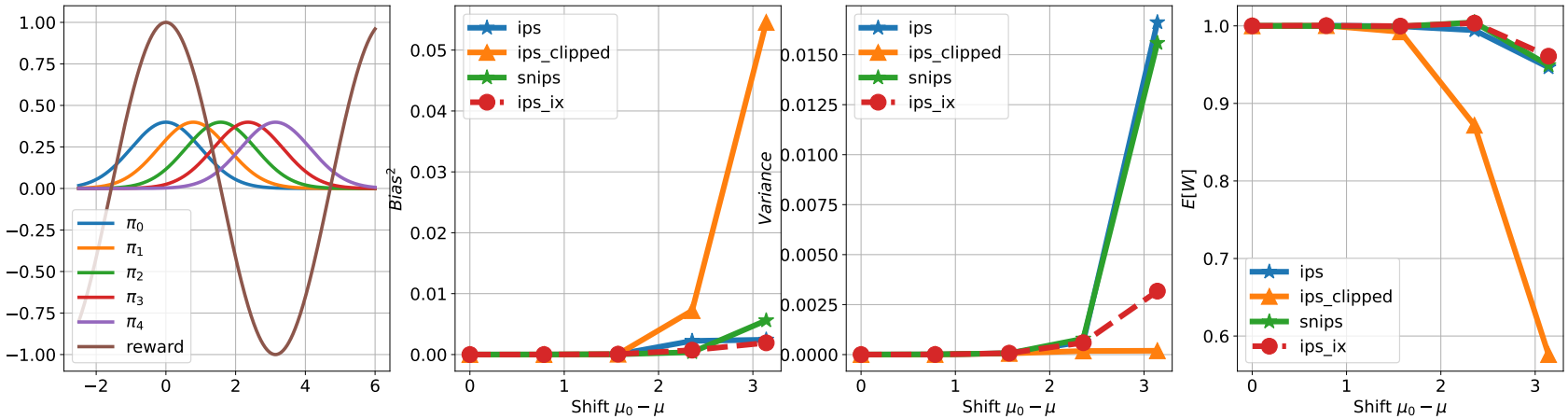


Figure: Setup (left), Bias (middle left), Variance (middle right), Average IPS weight (right). IPS-IX shows a low bias and compares favorably to IPS and SNIPS in terms of variance.

SCRM in practice

We consider discrete action real-world datasets (*Scene*, *Yeast*, *TMC2007*) and a continuous action setting (*Advertising*).

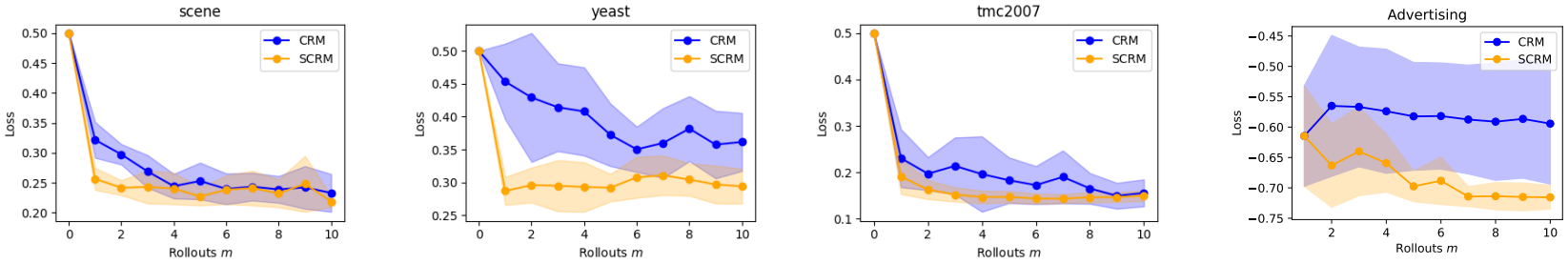


Figure: Test loss as a function of sample size on *Scene*, *Yeast*, *TMC2007*, *Advertising*, (from left to right). SCRM (in orange) converges faster and with less variance than CRM (in blue).

SCRM converges very fast, often within the first rollouts. Conversely, CRM needs more samples and the variance is higher.

Discussions and final worlds

We extended the CRM perspective for designing adaptive data collection experiments. We list the following future research directions:

- Develop guarantees to use all past samples, which is particularly effective in practice. Indeed, to avoid introducing dependencies in the excess risk bounds, we used geometric sample sizes to discard previous samples.

- Similar to online settings that involve an exploration-exploitation tradeoff, investigate the use of optimism in the face of uncertainty (OFUL) principle in SCRM, and investigate learning with safety constraints.

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